MIL-P-50686A (AR)
30 October 1979
SUPERSEDING
MIL-P-50686(MU)
15 April 1971

MILITARY SPECIFICATION

POWER SUPPLY: PP-4812/VPS-2; 10548440

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Power Supply, PP-4812/VPS-2, hereinafter referred to as a power supply, which is a part of Radar Set AN/VPS-2. The power supply provides the necessary voltages for functional operation of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-46357	Computer, Gun Direction: M18 Manufacturing Processes and Inspection of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

MIL-C-50329A	Converter, Electronic 10549013
MIL-R-50651	Radar Set AN/VPS-2: 10549058 (Range Only Radar)
MIL-A-50687	Antenna Assembly, AS-2049/VPS-2, 10548388

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F10548440	Power Supply, PP-4812/VPS-2
F10556312	Antenna Assembly AS-2049/VPS-2
F10549103	Base Support Assembly
C10554859	Toggle Switch

PACKAGING DATA SHEET

10548440	Packing of Power Supply PP-4812/VPS-2

(Copies of specifications, standards, drawings and packaging data sheets required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. The power supply shall be manufactured in accordance with Drawing F10548440 and drawings pertaining thereto. Subassemblies shall have met the requirements of the following specifications:

UNIT NO.	SUBASSEMBLY	SPECIFICATION
5A3	Converter, Electronic: 10549013	MIL-C-50329
5A1, 5A2	Servo, Electronics: 10548193	Not yet assigned

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

3.2 General specifications. Where applicable, the power supply shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishings
d. Part identification and marking
e. Workmanship
f. Soldering electrical connections

3.2.2 MIL-C-46357.

a. Etched circuit assemblies
b. Wiring
c. Threaded fasteners
d. Marking
e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The power supply shall be mounted on the vibration table in normal service position. The power supply shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for 2 hours along each of three mutually perpendicular axes within the frequency range of 10 to 40 hertz (Hz) and a half amplitude excursion of 1/64 inch. The rate of change of frequency at each frequency range shall not exceed 1 Hz per second. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The power supply shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of three mutually perpendicular axes. The shock impulses shall have an acceleration of 15 gravity units (G) and a pulse duration of 11 $\pm$ 1 milliseconds (ms). The G value of each impulse shall be within $\pm$ 10 percent when measured with a 0.2 to 250 Hz filter and the maximum G shall occur at approximately 5.5 ms.

3.3.3 Altitude.

3.3.3.1 Operating. The power supply shall meet the requirements of 3.4 and 3.5 during and after exposure to the barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

3.3.3.2 Nonoperating. The power supply shall meet the requirements of 3.4.and 3.5 after exposure for not less than 1 hour to the barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The power supply shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F. The adjusted output voltages shall be allowed to vary ± 5% at temperature extremes. The nominal adjusted maximum currents shall be allowed to increase 5% of the specified maximum values at temperature extremes.

3.3.4.2 Nonoperating (storage). The power supply shall not suffer permanent damage when subjected to not less than 4 hours at each temperature extreme of -65°F and +160°F, and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (nonoperating). The power supply shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity of 94 ± 4 percent for a period of not less than 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the power supply.

3.5 Performance. The power supply shall produce results as given in Table I when supplied the following inputs:

a. 24 ± 0.5 volts direct current (Vdc) between J1-A (+) and J1-B (-).

b. 115 volts alternating current (Vac), 400 hertz (Hz) between J3-s and J3-i (Ret).

3.5.1 Blower and control circuits.

3.5.1.1 Air flow. A fan shall be provided to pass airflow across the power supply converter electronics. Air shall be pulled in through the front louvered filter and exit out the rear louvers. The fan circuit shall have protection from overload using a 4-ampere (A) circuit breaker.

3.5.1.2 Time delay interlock. Shall be provided internal to the power supply. This time delay shall be 120 ± 12 seconds (s). When input power is applied as specified in 3.5, a delayed contact closure shall be provided between J5-Y (Ret) and J5-a (Radar operate).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

TABLE I. OUTPUTS

The power supply shall produce the following regulated output voltages when supplied with 24 ± 0.5 Vdc as specified in 3.5.

24 Vdc between J3-c(+), J3-t frame ground

24 Vdc between J5-G(+), J5H (Ret)

6.5 Vdc between J4-A(+), J4-V (Ret)

6.5 Vdc between J4-B(+), J4-C (Ret)

+15 Vdc between J4-x(+), J4-FF (Ret)

-15 Vdc between J4-y(-), J4-FF (Ret)

-15 Vdc between J5-M(-), J5-K (Ret)

+15 Vdc between J5-N(+), J5-K (Ret)

+23 Vdc between J4-Q(+), J4-CC (Ret)

-23 Vdc between J4-r(-), J4-CC (Ret)

+33 Vdc between J4-v(+), J4-w (Ret)

-33 Vdc between J4-c(-), J4-u (Ret)

-33 Vdc between J5-S(-), J5-U (Ret)

+29.3 Vdc between J4-AA(+), J4-j (Ret)

+29.3 Vdc between J5-V(+), J5W (Ret)

+29.3 Vdc between J5-X(+), J5Y (Ret)

3.5.1.3 Radar operational. Relay closure shall be provided between J2-P and J2-A before a "time out" signal is obtained and between J2-P and J2-R after "time out" signal is obtained.

3.5.1.4 Radiate command. When +24 Vdc is connected between J2-U (+) and J2-V (Ret), a Radiate Command (+24 Vdc) signal shall be present between J4-N (+) connected to J5-R (+) and J4-P (Ret).

3.5.2 Output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

3.5.2.1 Output voltages. The output voltages specified in Table I, shall be as specified in MIL-R-50329. The output voltage shall be protected from overload with a 35-A circuit breaker.

3.5.2.2 Current limiting. The current limiting features of the output voltages specified in Table I shall be as specified in MIL-R-50329.

3.5.2 3 Voltage regulation line and load. Regulation requirements of the output voltages shall be as specified in MIL-R-50329.

3.5.3 Servo electronics. The Servo electronics 10548193, housed as part of the power supply, shall be subjected to the Servo Drive Dynamic Test requirements of MIL-A-50687 (MU) after meeting the input-output requirements for functional operation basis, employing the output connectors of the power supply.

3.5.4 Subclutter visibility (SCV) test. The power supply after having met the requirements of 3.5.1 through 3.5.3 shall meet the subclutter visibility (SCV) requirements as specified in MIL-R-50651 when substituted into a Radar Set AN/VPS-2. Radar Set AN/VPS and test facilities shall be as specified in MIL-R-50651.

3.6 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.2 First article (initial production) submission. The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three power supplies manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certification of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certification for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) based in accordance with MIL-STD-105 and the sampling plans specified in 4.3.2.3 and 4.4 herein. Examinations and tests for packaging and marking shall be in accordance with

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

MIL-P-14232 and Section 5 herein. The tabulated classification of defects in 4.3.2.3 and 4.4 shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

4.3.2.3 Functional tests. The requirements and tests in Table II shall be inspected on a 100-percent basis. All examinations and tests shall be conducted at standard ambient temperatures ($60^{o}F$ to $90^{o}F$).

TABLE II. CLASSIFICATION OF DEFECTS
(100 percent inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR:		
101. Electrical	3.4	4.6.2
102. Performance	3.5	4.6.3
MINOR: NONE		

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling. One power supply shall be selected at random by a Government representative as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table III. Three power supplies shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in Table IV.

TABLE III. SPECIAL SAMPLING

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	Applicable drawings-Visual
302. General specifications	3.2	MIL-F-13926 & MIL-C-46357 - Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

TABLE IV. SPECIAL SAMPLING, ENVIRONMENTAL

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303. Vibration	3.3.1	4.6.1.1
304. Shock	3.3.2	4.6.1.2
305. Altitude	3.3.3	4.6.1.3
306. Temperature	3.3.4	4.6.1.4
307. Humidity (nonoperating)	3.3.5	4.6.1.5

4.4.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved (see 4.3.3).

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Test equipment. In conjunction with 4.5.2.2, the standard test equipment listed in Table V shall be utilized in performing the applicable tests of 4.6.

4.6 Test methods and procedures.

4.6.1 Environmental. Environmental tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in evnrionmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810. After performing the requirements of 4.6.2 and 4.6.3 at room ambient conditions, nominal settings shall be established with no further adjustments at environmental extremes. Voltage and current specifications shall be allowed to degrade per 3.3.4.1.

4.6.1.1 Vibration. This test shall be performed as specified in Procedure VIII, Method 514 of MIL-STD-810 as modified by paragraph 3.3.1 herein. After vibration, the power supply shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.1.

4.6.1.2 Shock. This test shall be performed as specified in Procedure I, Method 516 of MIL-STD-810 except that shock pulses shall be 11.0 ms and 15 G. After the shock impulses, the power supply shall be tested as specified in 4.6.2 and 4.6.3 to determine compliance with 3.3.2.

4.6.1.3 Altitude (operating and nonoperating. This test shall be performed as specified in Procedure I, Method 500 of MIL-STD-810 except that the operating altitude shall be limited to 10,000 feet (20.6 inches Hg). Operation of Step 4 shall be the performance tests specified in 4.6.3 herein. Nonoperating test of 40,000 feet (5.5 in. Hg) shall also be performed.

4.6.1.4 Temperature.

4.6.1.4.1 High (operating and nonoperating). This test shall be performed as specified in Procedure I, Method 501 of MIL-STD-810. The temperature of Step 3 shall be maintained for not less than 4 hours. The temperature of Step 4 shall be +155°F. In Step 5, the temperature shall be maintained for 4 hours and the power supply shall be tested in accordance with 4.6.2 and 4.6.3 during this time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

TABLE V. TEST EQUIPMENT LIST

NO.	ITEM	DESCRIPTION
1.	Power supply	Dc 22-30 V, 40 A, regulated to ± 0.5%.
2.	Voltmeter	Dc, 0 to 50 V, ± 0.1% accuracy, 20,000 ohms/volt or greater.
3.	Rms voltmeter	Frequency range: 10 Hz to 5 MHz. Meter accuracy: ± 5% Voltage range: 1 mV to 50 Vdc.
4.	Ammeter	Dc, 0 to 20 A in 3 ranges; accuracy, 1% of full scale.
5.	Test fixture	See Figure 1. See Table VI for test fixture parts list.
6.	Shaft clamp	See Figure 6.
7.	Servo director	Synchro transmitter compatible with 11CT4e. Simple harmonic motion of one cycle per 5 and 10 seconds with ±25° adjustable sector seam.
8.	Band pass filter	Frequency response: 0.02 Hz. to 20 kHz. Pass-band gain: 0dB ± 1dB. Attenuation slope: nominal 24dB per octave. Maximum attenuation: greater than 80dB. Input impedance: 22 megohms or greater. Parallel capacitance: 200 ± 50 pF.
9.	Recorder	Two channel. Sensitivity: 5mV/cm. to 20V/cm. Speed: 1.0 to 100 mm/s. Frequency response: 80 Hz.
10.	Power source, 400 Hz.	115V, 1A with distortion less than 1%.
11.	Base support assembly	In accordance with (ARRADCOM) drawing F10549103.
12.	Antenna Assembly	AS-2049/VPS-2: 10556312.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

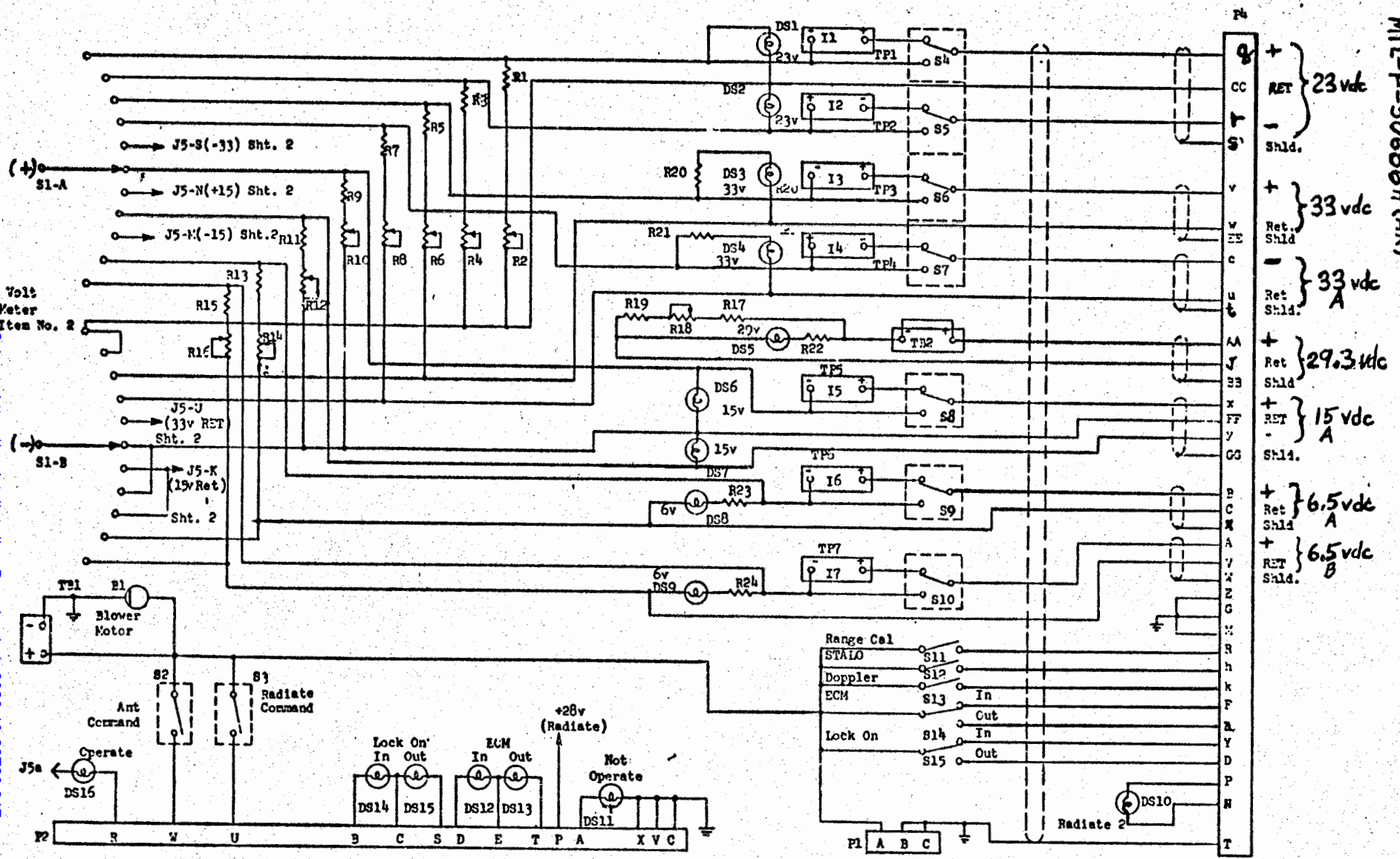


FIGURE 1. TEST FIXTURE (Sheet 1 of 2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

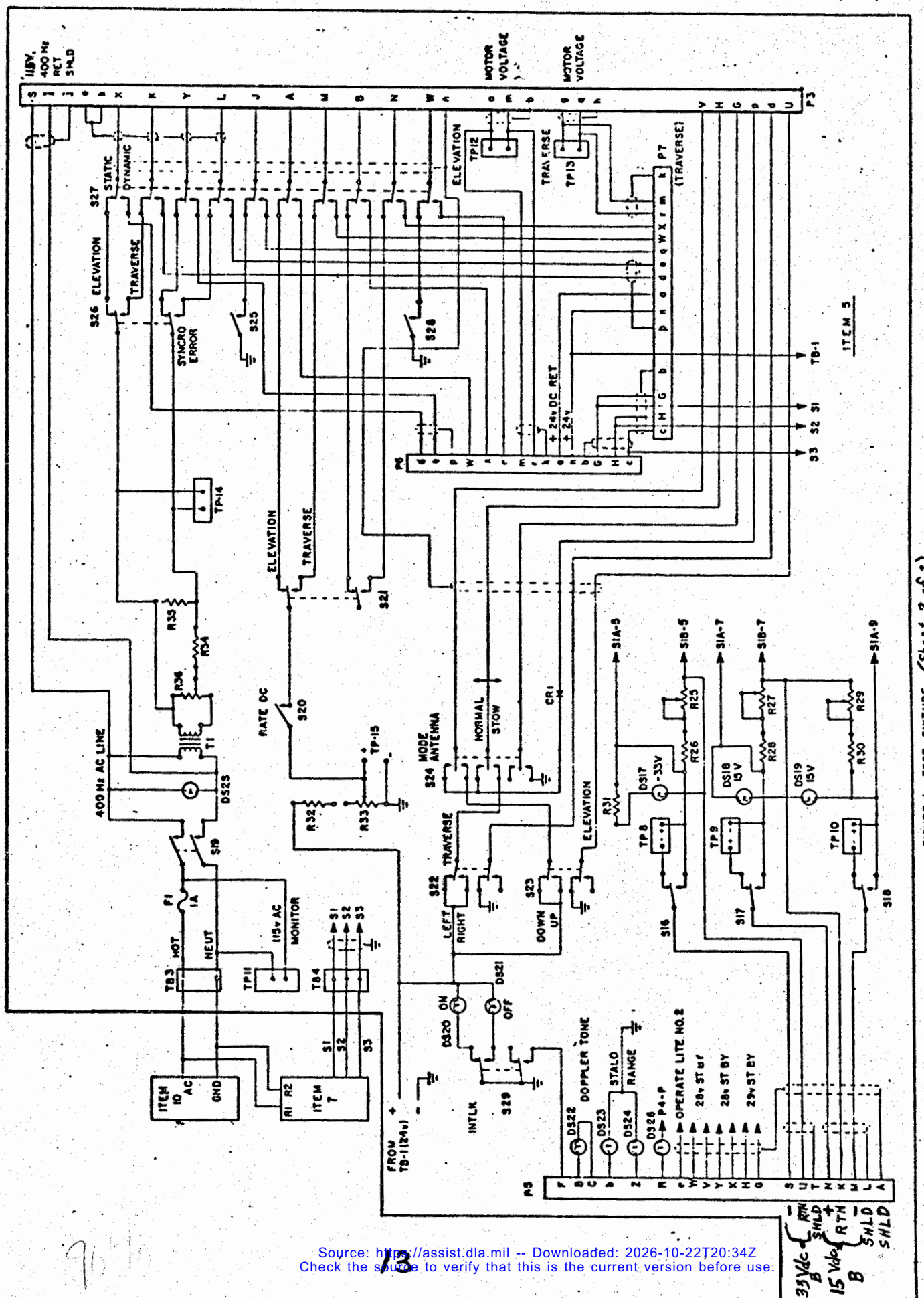


FIGURE 1. TEST FIXTURE (Sheet 2 of 2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

TABLE VI. TEST FIXTURE PARTS LIST

DESIGNATION	PART	DESCRIPTION
B1	Fan, cooling	100 CFM, Rotron Sentinel or equivalent
DS1 through DS5	Lamp and cartridge	24 V @ 40 mA
DS10 through DS17, DS20 through DS24		
DS6, DS7, DS18, DS19	Lamp and cartridge	14 V @ 80 mA
DS8, DS9	Lamp and cartridge	6 V @ 200 mA
DS25	Lamp and cartridge	115 Vac
F1	Fuse	1 A, type 3AG
P1	Connector	MS3106B16-10S
P2	Connector	L22TF21S6N-A
P3	Connector	L22TF41P6N-A
P4	Connector	L22TF55P6N-A
P5	Connector	PT06A20-26P (SR)
R1, R3	Resistor, fixed	18 ohms, 50 W
R5	Resistor, fixed	40 ohms, 50 W
R7, R26	Resistor, fixed	80 ohms, 50 W
R17, R19	Resistor, fixed	1.25 ohms, 500 W
R9, R11, R28, R30	Resistor, fixed	25 ohms, 25 W
R13, R15	Resistor, fixed	3 ohms, 50 W
R20, R21, R31	Resistor, fixed	RC32GF391K
R22	Resistor, fixed	RC20GF120K

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

TABLE VI. TEST FIXTURE PARTS LIST (CONT'D)

DESIGNATION	PART	DESCRIPTION
R23, R24	Resistor, fixed	RC20GF3R3K
R32, R35	Resistor, fixed	RC32GF242K
R34	Resistor, fixed	RC20GF240K
R2, R4	Resistor, variable wirewound	4 ohms, 50 W
R6	Resistor, variable, wirewound	8 ohms, 25 W
R8, R25	Resistor, variable, wirewound	25 ohms, 25 W
R18	Resistor, variable wirewound	1 ohm, 500 W
R10, R12, R27, R29	Resistor, variable wirewound	10 ohms, 12.5 W
R14, R16	Resistor, variable, wirewound	1 ohm, 10 W
R33	Resistor, variable, carbon	50 ohms, 2 W
R36	Resistor, variable, carbon	10 ohms, 2 W
S1	Switch, rotary	12 PDT, nonshorting
S2 through S18 S20, S25, S28	Switch, toggle	SPDT
S19, S21 through S23, S26, S29	Switch, toggle	DPDT
S24	Switch, toggle	4 PDT
S27	Switch, toggle	C10554859

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

TABLE VI. TEST FIXTURE PARTS LIST (CONT'D)

DESIGNATION	PART	DESCRIPTION
T1	Transformer	6.3 Vac, 0.6 A. Triad F13X or equivalent
TB1, TB2, TB3, TB4	Terminal	MIL type 38TB3
TP1 through TP14	Jack, tip	E.F. Johnson type 105
XF1	Fuseholder	Littelfuse, 342001 or equivalent

4.6.1.4.2 Low (operating and nonoperating). This test shall be performed as specified in Procedure I, Method 502 of MIL-STD-810. The temperature of Step 2 shall be -65°F maintained for not less than 4 hours. The temperature of Step 4 shall be -40°F. In Steps 5 and 7, the power supply shall be tested in accordance with 4.6.2 and 4.6.3.

4.6.1.5 Humidity (nonoperating). This test shall be performed in accordance with Procedure II, Method 507 of MIL-STD-810. The power supply shall be tested in accordance with 4.6.3.

4.6.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.6.3 Performance. The selected power supplies shall be tested as follows: connect test harness, Item 5, as shown in Figure 2.

4.6.3.1 Initial conditions.

4.6.3.1.1 Position power supply controls as follows:

a. Set Converter switch CB1 to OFF
b. Set Traverse switch CB2 to OFF
c. Set Elevation switch CB3 to OFF
d. Set Control Circuit switch CB4 to OFF
e. Set Blowers switch CB5 to OFF
f. Set 6.5 Vdc switch CB6 to OFF
g. Set S27 to Static Position

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.6.3.1.2 Set test fixture, Item 5, controls as follows:

a. +6.5V A switch to OUT
b. -15V A switch to OUT
c. +15V A switch to OUT
d. -15V B switch to OUT
e. +15V B switch to OUT
f. -33V A switch to OUT
g. -33V B switch to OUT
h. +6.5V B switch to OUT
i. +23V switch to OUT
j. -23V switch to OUT
k. +33V switch to OUT
l. INTERLOCK switch to OPEN
m. ANTENNA COMMAND switch to OFF
n. ERROR ADJUST control fully counterclockwise (ccw)
o. RATE ADJUST control fully counterclockwise
p. LOCK ON switch to OUT
q. ECM switch to OUT
r. STALO CALIB switch to OFF
s. RANGE CALIB switch to OFF
t. RADIATE COMMAND switch to OFF
u. 115 Vac switch to OFF
v. ERROR switch to ELEVATION
w. DOPPLER TONE switch to OFF
x. RATE switch to OFF
y. RATE EL/TRAV to ELEVATION
z. ANTENNA MODE switch to OFF (center position)
aa. ELEVATION switch to DOWN
bb. TRAVERSE switch to LEFT

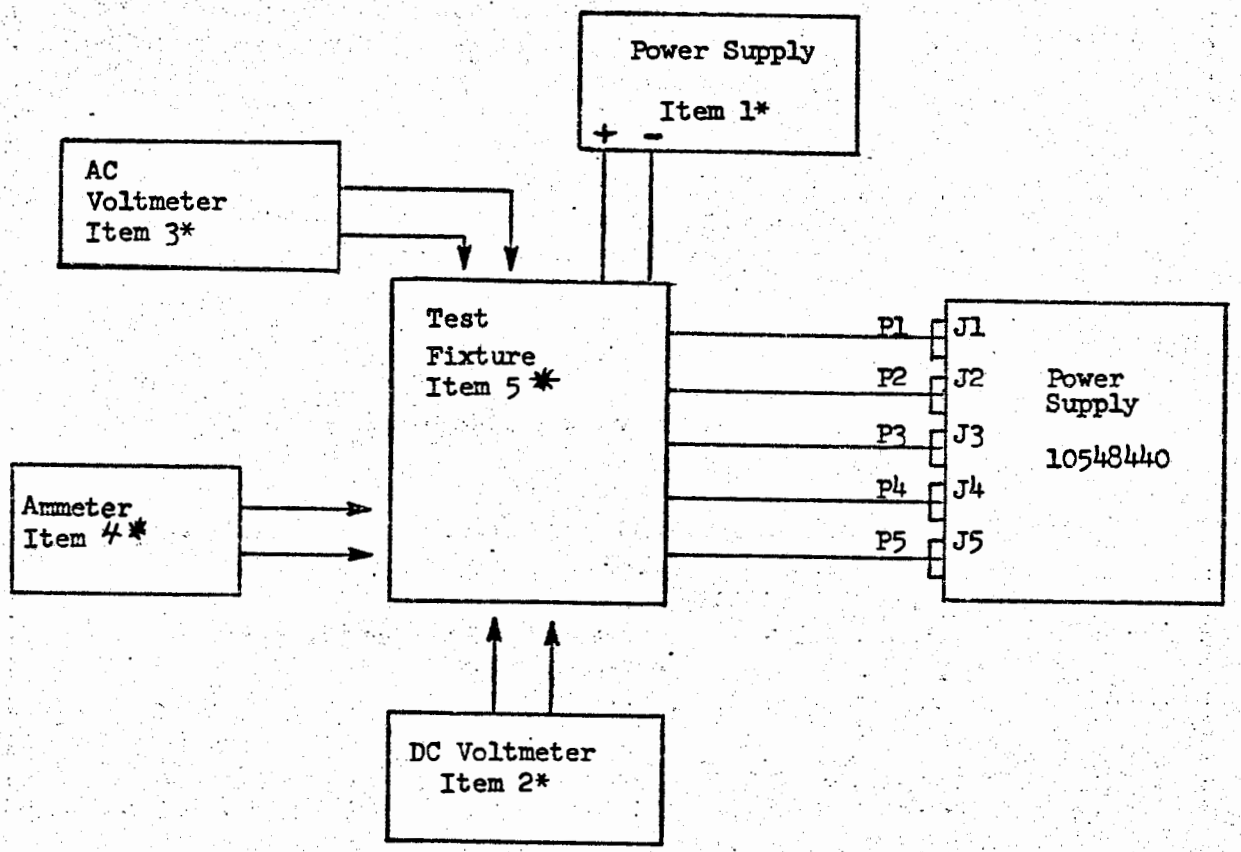
4.6.3.1.3 Set power supply, Item 1, voltage control ccw. Operate power switch to ON. Connect voltmeter, Item 2, to the test fixture of the power supply and adjust the power supply to obtain 24 $\pm$ 0.5 Vdc.

4.6.3.2 Continuity and blower operation.

4.6.3.2.1 Operate power supply, Item 1, Power switch to ON. Observe that Interlock Open and Lock-On Out lamps are lit on test fixture, Item 5.

4.6.3.2.2 Operate Blower switch on power supply, unit 5, to ON. Observe and record that blower in unit 5 operates and note direction of air flow to be entering the front of the unit and exiting out the rear of the unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.



*Refer to 4.5.2.3

FIGURE 2. UNIT 5 TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.6.3.3 Power supply (unit 5).

4.6.3.3.1 Operate the Control circuit switch on power supply, unit 5, to ON. Operate the Converter switch on power supply, unit 5, to ON. Observe that the pilot lamps for the following voltages are lit: +23V, -23V, +33V, -33V A, -33V B, +15V A, +15V B, -15V A, -15V B.

4.6.3.3.2 Connect Item 2 voltmeter's red-positive and black-negative leads to the respective red and black dc voltage test jacks on the test fixture.

4.6.3.3.3 Operate the voltage selector switch on test fixture, Item 5, and observe the voltages as listed:

+23 ± 1V
-23 ± 1V
+33 ± 1V
-33 ± 1V A
-33 ± 1V B
+15 ± 1V A
+15 ± 1V B
-15 ± 1V A
-15 ± 1V B

a. Operate 6.5 Vdc switch on radar power supply to ON.
b. Observe that the +6.5 Vdc A and +6.5 Vdc B pilot lamps on test fixture, Item 5, are lit.
c. Operate voltage select switch on test fixture to +6.5V A and +6.5V B.
d. Observe that the voltages as indicated on the voltmeter are within the following limits:

+6.5 ± 1V A
+6.5 ± 1V B

4.6.3.4 Load currents. Set ammeter, Item 4, to the 10A scale; Connect the ammeter to the circuit test jacks beginning at 6.5V A. Operate 6.5V A ammeter in-circuit-switch to up and monitor the current (operate Item 4 as required to obtain a high on-scale reading). Repeat above steps for +23V, -23V, +33V A, -33V A, -33V B, +15V A, +15V B, -15V A, -15V B positions. Remove the jumper on the 29.3 V terminal board and monitor a current of 10 A. Adjust if required. The following currents should be present and adjusted to obtain the values specified below:

VOLTAGE MONITORED	CURRENT REQUIRED
29.3V	9.5 to 10.5 A
+33V	700 to 800 mA
-33V A&B	350 to 400 mA
+15V A & B	475 to 525 mA
-15V A&B	475 to 525 mA
6.5V A & B	1.65 to 1.85
+23V	1 to 1.4 A
-23V	1 to 1.4 A

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.6.3.5 Output voltages. Operate the voltage selector switch; observe and record the voltages as listed below:

SWITCH POSITION	NOT LESS THAN (Vdc)	NOT MORE THAN (Vdc
-15V A	-14.9	-15.1
+15V A	+14.9	+15.1
+6.5V A	+6.0	+6.7
+6.5V B	+6.0	+6.7
-15V B	-14.9	-15.1
+15V B	+14.9	+15.1
-33V A	-32.7	-33.3
-33V B	-32.7	-33.3
+33V	+32.7	+33.3
-23V	-22.7	-23.3
+23V	+22.7	+23.3

4.6.3.6 Control circuits.

4.6.3.6.1 Operate Interlock switch on test fixture, Item 5, to CLOSED. Observe and record that INTERLOCK CLOSED lamp is lit. Observe and record that operate lamp lights between 108 and 132 seconds after interlock lamp is lit.

4.6.3.6.2 Utilizing voltmeter, Item 2, and test jacks on test fixture, Item 5, observe that the following voltages are within limits and of proper polarity:

+24 V standby between 20 and 30 Vdc
+29 V standby between 28.7 and 29.5 Vdc
+29 V operate between 28.7 and 29.5 Vdc

4.6.3.6.3 Operate LOCK ON switch on test fixture, Item 5, to IN. Observe and record that LOCK ON IN lamp is lit and LOCK ON OUT lamp is out.

4.6.3.6.4 Operate LOCK ON switch on test fixture, Item 5, to OUT. Observe and record that LOCK ON IN lamp is out and LOCK ON OUT lamp is lit.

4.6.3.6.5 Operate ECM switch on test fixture to IN. Observe and record that ECM IN lamp is lit and ECM OUT lamp is out.

4.6.3.6.6 Operate ECM switch on test fixture to OUT. Observe that ECM IN lamp is out and ECM OUT lamp is lit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.6.3.6.7 Operate STALO CALIB switch on test fixture to ON. Observe that the STALO lamp on the test fixture is lit. Operate STALO CALIB switch on test fixture to OFF.

4.6.3.6.8 Operate RANGE CALIB switch on test fixture, Item 5, to ON. Observe that RANGE lamp on test fixture is lit. Operate RANGE CALIB switch on test fixture to OFF.

4.6.3.6.9 Operate RADIATE COMMAND switch on test fixture, Item 5, to ON. Observe that radiate lamps no. 1 and no. 2 are lit. Operate RADIATE COMMAND switch on test fixture to OFF. Operate Converter and 6.5 Vdc switches on radar power supply to OFF.

4.6.4 Servo functional test (elev).

4.6.4.1 Operate Traverse, Elevation circuit breakers on radar power supply to ON. Operate power supply line switch to OFF. Connect the root mean square (rms) voltmeter, Item 3, to the Synchro Error test jacks on test fixture, Item 5. Connect the voltmeter to the Elevation test jacks on test fixture, Item 5. (Red to + of voltmeter). Operate 115 Vac switch on test fixture, Item 5, to ON. Observe that 115 Vac pilot lamp is lit on test fixture.

4.6.4.2 Operate ANTENNA COMMAND switch to ON. Operate AC LINE switch on test fixture to ON. Operate ANTENNA MODE switch to NORMAL. Adjust ERROR ADJUST to obtain between -10.0 to -10.5V on the voltmeter, Item 2. Observe that rms voltmeter reads between 9.2 and 14 mV.

4.6.4.3 Operate ELEV LIMIT switch to ON. Rotate the ERROR ADJUST clockwise (cw) to obtain between -10.0 and -10.5 Vdc on voltmeter, Item 2. Observe and record that rms voltmeter reads between 117 and 176 mV.

4.6.4.4 Return ERROR ADJUST control fully counterclockwise. Operate ELEV LIMIT switch on test harness to OFF. Operate ANTENNA MODE switch on test harness to STOW. Observe that voltmeter, Item 2, reads greater than +22.5 V.

4.6.4.5 Operate ELEVATION switch on test fixture, Item 5, to UP. Observe and record that voltmeter, Item 2, reads greater than -22.5 V.

4.6.4.6 Operate ELEVATION switch on test fixture, Item 5, to DOWN. Operate ANTENNA MODE switch on test fixture to NORMAL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

Operate RATE DCV switch on test fixture to ON. Rotate RATE ADJUST on test fixture to produce -10.0 to -10.5 Vdc on voltmeter, Item 2. Move leads of voltmeter, Item 2, to Rate test points. Observe dc voltage is between +0.67 and +1.00 Vdc.

4.6.5 Servo functional test (traverse).

4.6.5.1 Operate ANTENNA MODE switch to OFF (center). Operate RATE DCV switch on test fixture to OFF. Adjust RATE adjust fully counterclockwise. Operate SYNCHRO ERROR switch to traverse. Connect the voltmeter to TRAVERSE test jacks. Operate antenna mode switch to NORMAL. Adjust ERROR ADJUST to obtain between -10.0 and -10.5 volts on voltmeter, Item 2.

4.6.5.2 Observe that the rms voltmeter, Item 3, reads between 9.2 and 14 mV.

4.6.5.3 Operate TRAVERSE limits switch to ON. Rotate the ERROR adjust clockwise to obtain between -10.0 and -10.5 Vdc on voltmeter, Item 2. Observe and record that root mean square (rms) voltmeter reads between 117 and 176 millivolts (mV).

4.6.5.4 Return ERROR adjust control fully ccw. Operate TRAVERSE limits switch on test fixture, Item 5, to OFF. Operate ANTENNA MODE switch on test fixture to STOW. Observe that voltmeter, Item 2, reads greater than +22.5 V.

4.6.5.5 Operate TRAVERSE switch on test fixture to RIGHT. Observe that voltmeter, Item 2, reads greater than -22.5 V.

4.6.5.6 Adjust the power supply for 24.0 V. Operate TRAVERSE switch on test fixture to left and RATE EL/TRAV switch to TRAVERSE. Operate ANTENNA MODE switch on test fixture, Item 5, to NORMAL. Operate RATE DC switch on test fixture to ON. Rotate RATE ADJUST on test fixture to produce -10.0 to -10.5 Vdc on voltmeter, Item 2. Move voltmeter leads to Rate test points. Observe that the meter reading is between +0.67 and +1.00 Vdc.

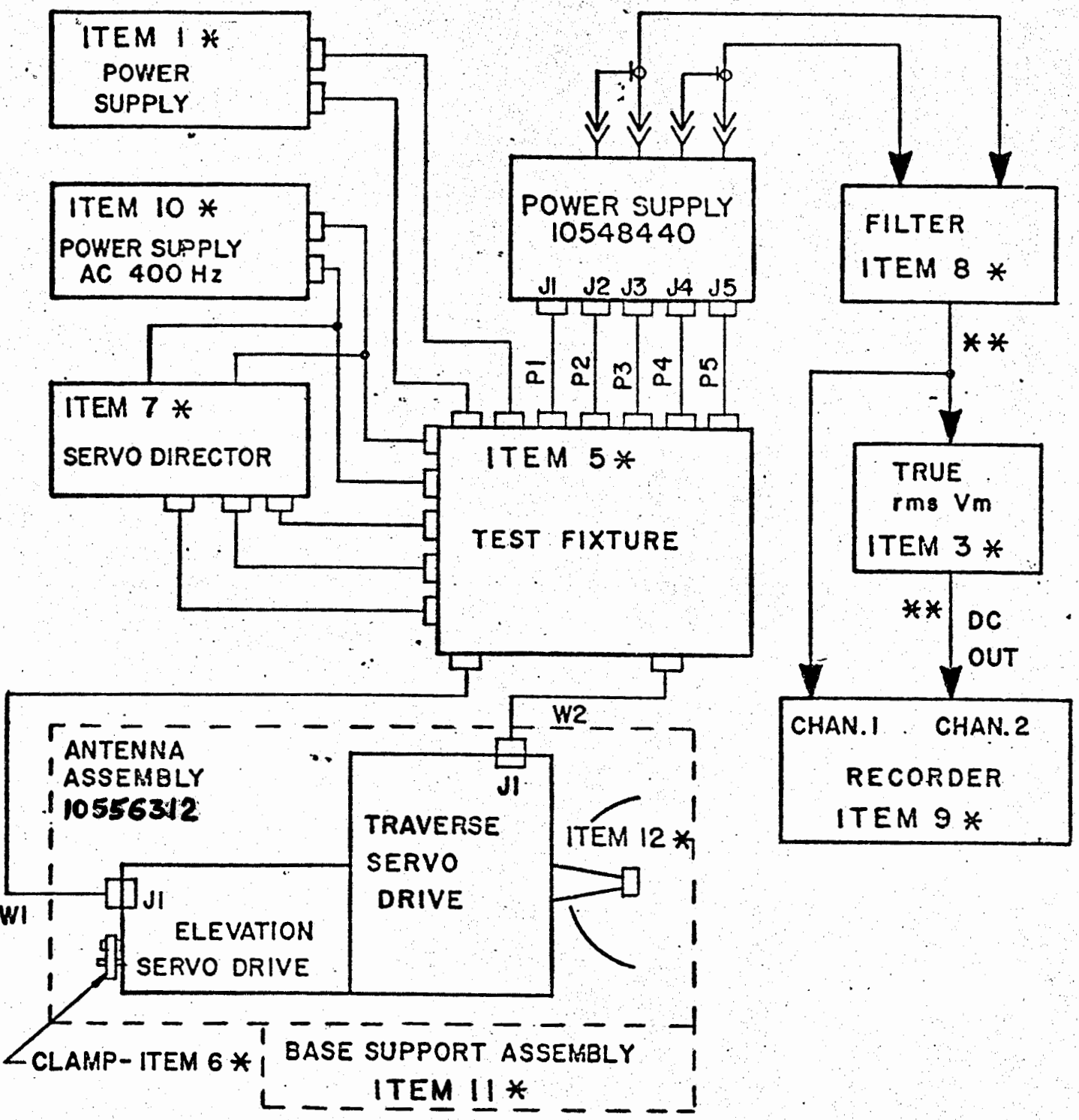
4.6.5.7 Servo drive dynamic test. After performing the tests of 4.6.4 through 4.6.5.6, connect the power supply per Figure 3 and perform the servo drive dynamic test as specified in specification MIL-A-50687 (AR). S27 of Figure 1 shall be in the dynamic position.

4.6.5.8 Operate ANTENNA MODE switch to OFF. Operate 115 Vac switch to OFF. Operate ac line switch to OFF. Remove radar power supply UNIT 5 from test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

4.6.6 Subclutter visibility (SCV) test. After performing the tests of 4.6.3 through 4.6.5.7, substitute the power supply into a Radar Set AN/VPS-2 and perform the subclutter visibility (SCV) test as specified in MIL-R-50651. The Radar Set AN/VPS-2 and test facilities used shall be as specified in MIL-R-50651.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.



* REFER TO 4.5.2.3 OF MIL-A-50687

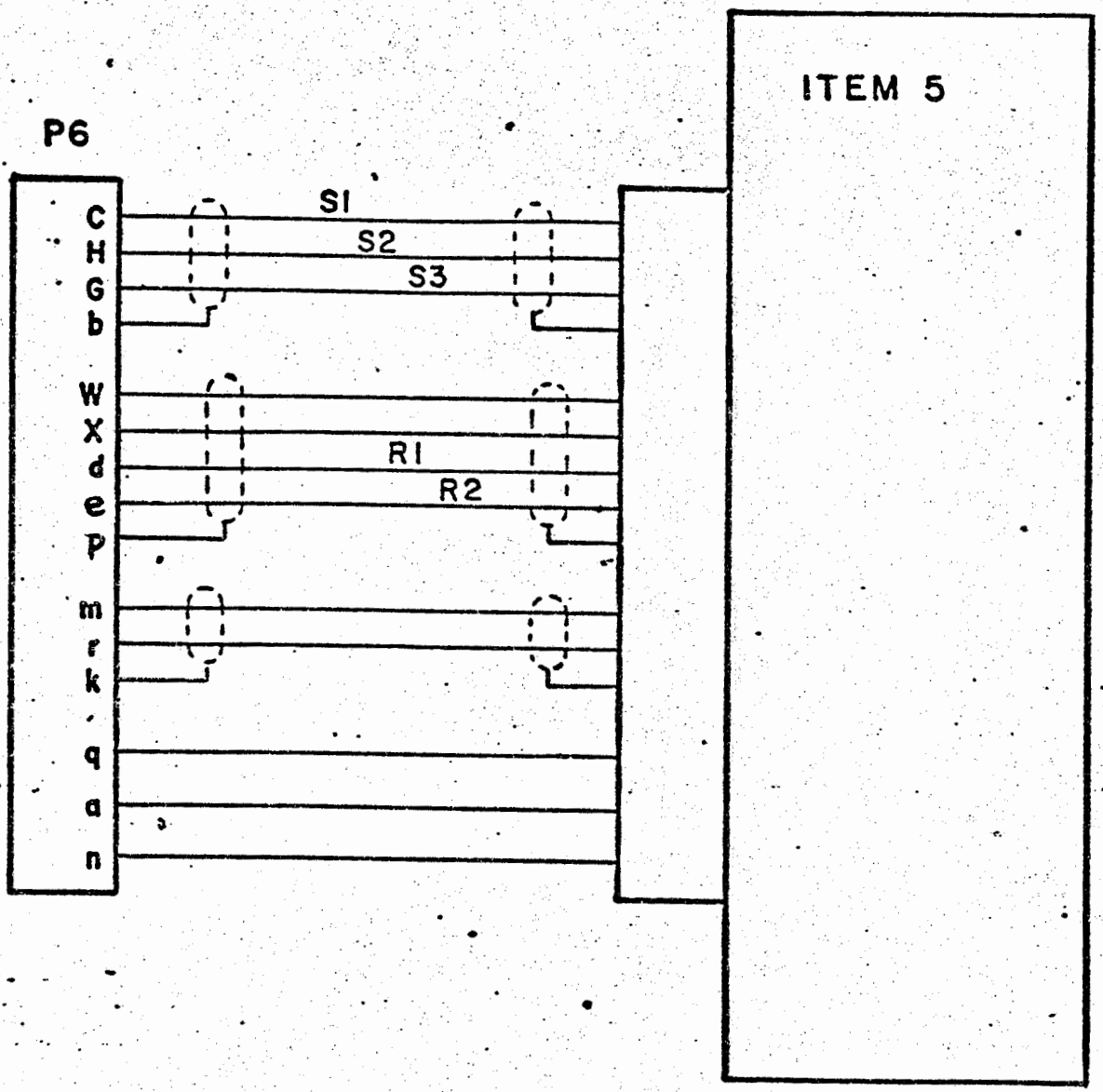
** USE SOLID CONNECTIONS FOR ITEMS 8, 3, 9, TRAVERSE & ELEVATION TESTS

W1- SEE FIGURE 4

W2- SEE FIGURE 5

FIGURE 3 TEST SETUP

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

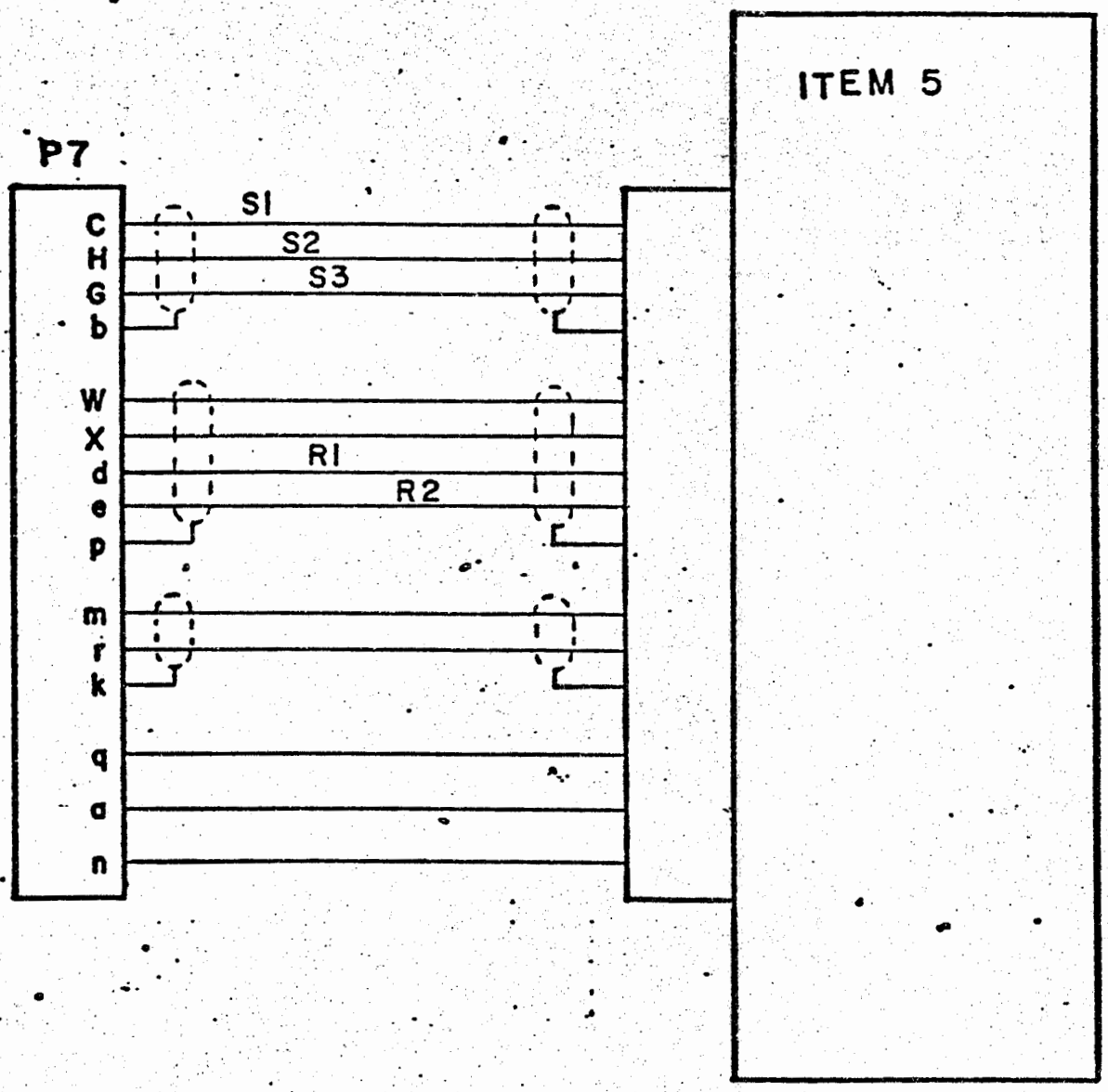


CONDUCTOR SIZE 16 AWG, ALL OTHERS 22 AWG.

P6—CONNECTOR, PLUG FRANKFORD ARSENAL PART NO. 10549134-8

FIGURE 4 TEST CABLE W1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.



CONDUCTOR SIZE 16AWG, ALL OTHERS 22 AWG

P7-CONNECTOR, PLUG FRANKFORD ARSENAL PART NO. 10549134-7

FIGURE 5 TEST CABLE W2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

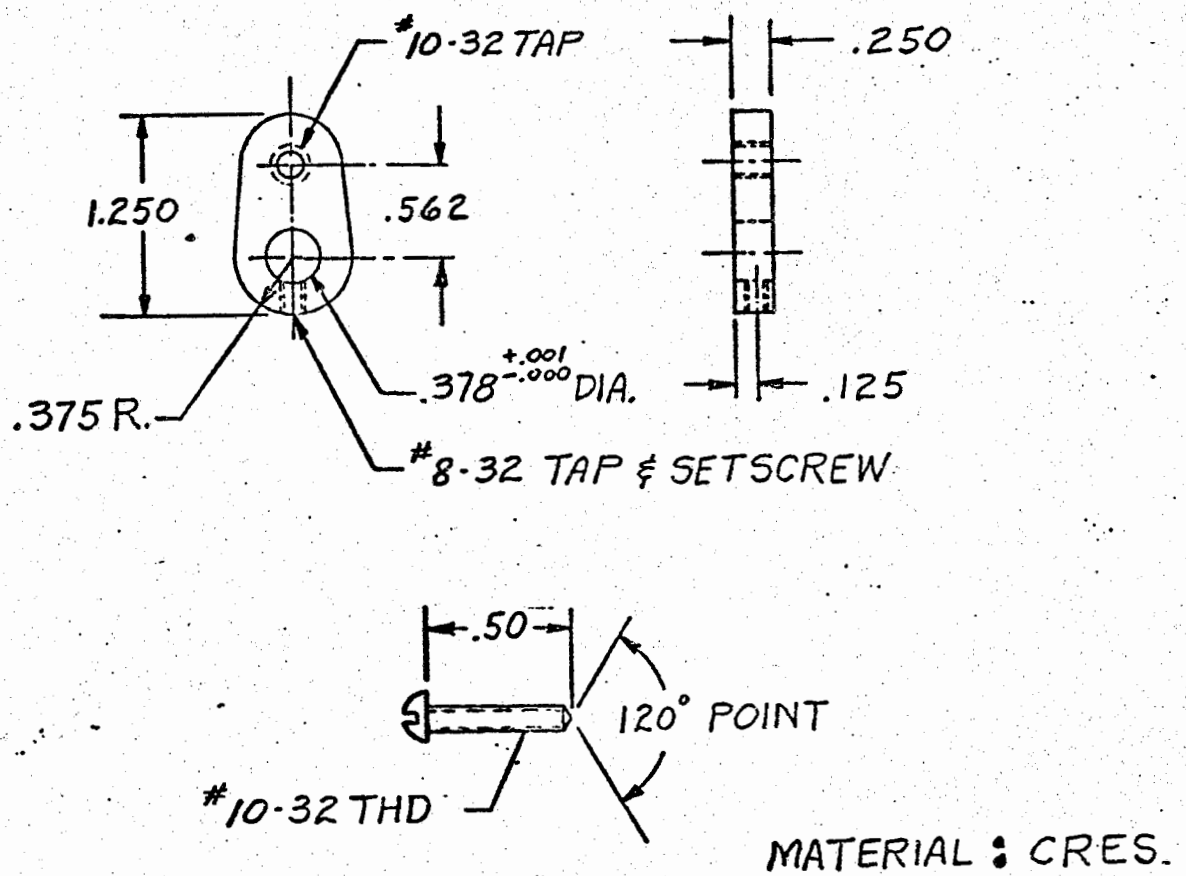


FIGURE 6. SHAFT CLAMP

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10548440. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The power supply provides all the low voltages required to functionally operate the Radar Set AN/VPS-2. The servo amplifiers are housed as part of the power supply and are used to direct the servo drive controls to position the antenna.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
ARMY-AR

Preparing Acitvity:
ARMY-AR

Project NO. 1285- A494

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-50686A Power Supply: PP-4812/VPS-2, 10548440

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:34Z
Check the source to verify that this is the current version before use.